

REAL SOURCE DOCUMENT — MIT COURSE CATALOG

Institution: Massachusetts Institute of Technology (MIT)

Course: 6.1210 — Introduction to Algorithms (formerly 6.006)

Source URL: <https://catalog.mit.edu/search/?P=6.1210>

Supporting source: <https://catalog.mit.edu/subjects/6/> (MIT Course 6 subject listing)

Date Accessed: March 29, 2025

Department: Electrical Engineering and Computer Science (EECS)

COURSE DESCRIPTION: 6.1210 Introduction to Algorithms

OFFICIAL CATALOG ENTRY:

Prereq: 6.100A and (6.1200[J] or (6.120A and (6.3700, 6.3800, 18.05, or 18.600)))

Level: U (Undergraduate)

Offered: Fall, Spring

Units: 5-0-7 (12 units total) [lecture-recitation-independent study]

Enrollment may be limited.

Instructors: E. Demaine, S. Devadas (representative; may vary by term)

DESCRIPTION:

Introduction to mathematical modeling of computational problems, as well as common algorithms, algorithmic paradigms, and data structures used to solve these problems. Emphasizes the relationship between algorithms and programming, and introduces basic performance measures and analysis techniques for these problems.

TOPICS COVERED (from course materials and catalog):

- Algorithmic thinking and problem modeling
- Sorting algorithms and their analysis
- Search trees, heaps, and hashing
- Divide-and-conquer paradigm
- Dynamic programming
- Greedy algorithms
- Graph algorithms (BFS, DFS, shortest paths)
- Basic performance measures (Big-O, Big-Theta, Big-Omega)
- Introduction to NP-completeness

PREREQUISITE DETAILS:

The prerequisite structure for 6.1210 is:

- 1. 6.100A (Introduction to Computer Science Programming in Python) — must be completed**
- 2. PLUS one of the following mathematics options:**

- a. 6.1200[J] (Mathematics for Computer Science), OR
- b. 6.120A (Discrete Mathematics, older version) AND one probability course: 6.3700, 6.3800, 18.05, or 18.600

Students who do not have 6.1200[J] must have both the older discrete math prerequisite AND a probability course. This is an either/or structure, not both required.

NOTE ON LEGACY NUMBERING:

6.1210 was formerly numbered 6.006 in the pre-2022 curriculum. Students who completed 6.006 have satisfied the equivalent of 6.1210 for prerequisite purposes. The catalog explicitly notes this equivalency for students in the curriculum transition.

DOWNSTREAM COURSES REQUIRING 6.1210:

- 6.1220[J] Design and Analysis of Algorithms: Prereq 6.1200[J] and 6.1210
- 6.1400[J] Computability and Complexity Theory: Prereq 6.1200[J] and 6.1210
- 6.1800 Computer Systems Engineering: Prereq 6.1210 and 6.1910
- 6.5081 Performance Engineering of Software Systems: Prereq 6.1020, 6.1210, and 6.1910
- Various 6.3xxx and 6.4xxx AI/ML courses require 6.1210 as part of their prerequisite chain

GRADING AND FORMAT:

6.1210 includes both lecture and problem sets. Enrollment may be limited in some semesters; students should register early. The course uses Python for programming assignments.